

From Seed to Seed: The Life of a Bean Plant

An informational passage for the Plant Life Cycles unit (Lessons 1 and 7).

A bean plant starts as a small, hard seed. It may not look alive, but it is. Inside the seed is a tiny baby plant. The seed also holds food for the baby plant. A tough seed coat covers the outside and keeps it safe.

A seed can wait a long time. To start growing, it needs water, air, and warmth. When a bean is planted in damp, warm soil, it soaks up water and swells. Then the seed coat splits open. The seed begins to **germinate**. That means it starts to grow.

First, a root pokes out of the seed and grows down into the soil. Roots hold the plant in place. They also soak up water. Next, a stem pushes up through the soil. Soon a little green plant pops out of the ground. Now it is called a **seedling**.

By now, the seedling has used up its stored food. Its green leaves make new food from sunlight, air, and water. The stem grows taller. It carries water from the roots up to the leaves. In a few weeks, the seedling grows into an adult plant.

An adult bean plant grows flowers. Each flower makes a yellow dust called **pollen**. Moving pollen so a flower can make seeds is called **pollination**. A bean flower can pollinate itself. Bees help too. They visit flowers to drink sweet nectar, and pollen sticks to their fuzzy bodies. When a bee flies to the next flower, some pollen rubs off.

After a flower is pollinated, it changes. The petals dry up and fall off. A long green pod grows in its place. The pod is the bean plant's **fruit**. A fruit is the part of a plant that holds its seeds. Inside the pod, new seeds grow bigger every day.

At the end of summer, the plant turns brown and dies. The pods dry out and split open. Some seeds fall to the ground. A gardener might save others in a jar. Next spring, one of those seeds can be planted. It will germinate, grow, flower, and make new seeds. This pattern is called a **life cycle**. It goes from seed to seed, again and again.

Try it: Grow a bean in a cup

1. Line a clear cup with a wet paper towel.
2. Stuff another wet towel in the middle.
3. Slide a bean between the towel and cup.
4. Keep it damp. Look and draw every day.

Which comes out first, the root or the stem?

Think about the text

1. What does a bean seed need before it can germinate? Copy the sentence that tells you.
2. How does a bee help a bean plant make new seeds? Use two details from the passage.
3. The title says a bean plant's life goes "from seed to seed." What does that mean? Use details from the beginning and end of the passage.